

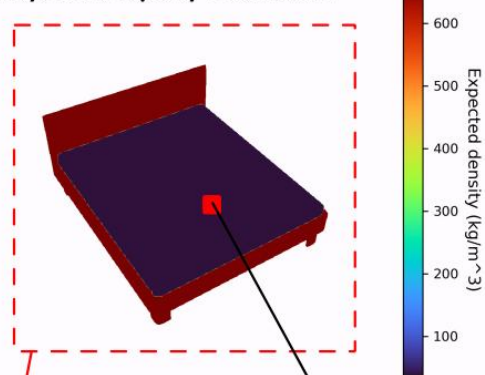
01



Physical Property Estimation

Infer material and mechanical properties of objects, including appearance, mass, rigidity and deformability, from visual observations.

Physical Property Estimation



Per-point* density

Full object mass

35.1 kg/m³

64.0 kg

02



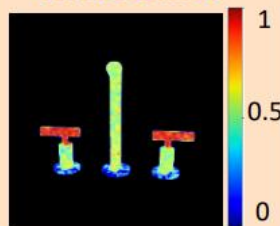
Affordance Reasoning

Reason actionable possibilities of objects, evolving from static geometry to human demonstration and active interactive learning.

Absolute Scale

Physical dimension:
27.51×19.8×6.76
cm

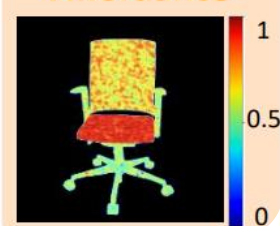
Affordance



Absolute Scale

Physical dimension:
98.92×69.2×64.3
cm

Affordance



03



Physical Causal Reasoning

Predict actions outcome, temporal dynamics and counterfactual physical events.



04



Physical Consistency Verification

Evaluate physical plausibility and validate executable simulation assets beyond visual realism.

Qwen3-VL-30B-A3B-Instruct

Reasoning: The wooden calendar is (✓ spatial perception) located inside the cabinet, on a lower shelf. The car is positioned on the floor near the cabinet, ..., its movement is (✗ inaccurate indication) constrained by the floor and the cabinet's structure. (✗ wrong causality) Therefore, there is no indication that the car's movement would interfere with accessing the calendar. Therefore, it is possible to remove the calendar without being blocked by the car.

Answer: (A) Yes

With a later frame as input:

The toy car is (✓ static perception) positioned immediately in front of the blocks, leaving no forward clearance for the blocks to be slid outward. (✓ correct causality) Because the blocks cannot move past the car and the cabinet sides restrict lateral movement, there is no unobstructed path for removal.

Answer: (B) No

Question:

If the car moves backward, can the wooden calendar be removed from the cabinet without being blocked by the car?

Answer: by (A) Yes, (B) No or (C) Not sure.

Initial Frame



Later Frame

